

MCQs on Data Communication (Class XII – Computer Science)

Chapter Overview & Introduction

1. The chapter discussed deals primarily with which subject?
(A) Computer Architecture
(B) Data Communication
(C) Operating Systems
(D) Software Engineering
 2. Which of the following is **NOT** listed in the chapter contents?
(A) Concept of Communication
(B) Transmission Media
(C) Database Normalization
(D) Protocol
 3. The quote by Yuval Noah Harari in the chapter emphasizes:
(A) Mechanical automation
(B) Human–smartphone integration
(C) Space technology
(D) Internet security
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Concept of Data Communication

4. The term “Data Communication” consists of which two words?
(A) Data and Network
(B) Signal and Media
(C) Data and Communication
(D) Information and Transfer
5. Data can be in the form of:
(A) Text only
(B) Image and audio only
(C) Video only
(D) Text, image, audio, video, multimedia
6. Communication refers to:
(A) Storing data
(B) Processing data
(C) Sending or receiving data
(D) Encrypting data
7. Data communication refers to the exchange of data between:
(A) Only computers
(B) Two or more connected devices
(C) Only servers
(D) Only mobile phones

8. Devices involved in data communication must be capable of:
 - (A) Processing data only
 - (B) Storing data only
 - (C) Sending and receiving data
 - (D) Encrypting data
 9. Which of the following is an example of a communication device?
 - (A) Scanner
 - (B) Printer
 - (C) Mobile phone
 - (D) All of the above
 10. In a network, devices such as computer, printer, server and switch are connected using:
 - (A) Processor
 - (B) Media
 - (C) Software
 - (D) Memory
-

Components of Data Communication

11. How many main components are involved in data communication?
 - (A) Three
 - (B) Four
 - (C) Five
 - (D) Six
12. Which of the following is **NOT** a component of data communication?
 - (A) Sender
 - (B) Receiver
 - (C) Compiler
 - (D) Protocol
13. The communication medium is also known as:
 - (A) Channel
 - (B) Protocol
 - (C) Node
 - (D) Message
14. A sender is:
 - (A) A device that receives data
 - (B) A device that stores data
 - (C) A device that sends data
 - (D) A device that processes data
15. Which of the following can act as a sender?
 - (A) Smartwatch
 - (B) Walkie-talkie
 - (C) Mobile phone
 - (D) All of the above

16. A receiver is a device that:
- (A) Sends data
 - (B) Receives data
 - (C) Encrypts data
 - (D) Compresses data
17. In computer networks, sender and receiver are called:
- (A) Signals
 - (B) Channels
 - (C) Nodes
 - (D) Packets
18. Message in data communication refers to:
- (A) The protocol
 - (B) The medium
 - (C) The data being exchanged
 - (D) The hardware
19. Which of the following is a valid form of a message?
- (A) Audio
 - (B) Video
 - (C) Image
 - (D) All of the above
20. Communication media can be:
- (A) Wired only
 - (B) Wireless only
 - (C) Wired or wireless
 - (D) Optical only
21. Ethernet and HTTP are examples of:
- (A) Media
 - (B) Protocols
 - (C) Signals
 - (D) Nodes

Measuring Capacity of Communication Media

22. Transmission medium is also known as:
- (A) Node
 - (B) Channel
 - (C) Protocol
 - (D) Message
23. The capacity of a channel refers to:
- (A) Length of cable
 - (B) Number of devices
 - (C) Maximum traffic it can carry
 - (D) Signal strength

24. Channel capacity is measured in terms of:
- (A) Frequency only
 - (B) Bandwidth and data transfer rate
 - (C) Voltage
 - (D) Amplitude
-

Bandwidth

25. Bandwidth refers to:
- (A) Signal power
 - (B) Range of frequencies
 - (C) Number of packets
 - (D) Transmission delay
26. Higher bandwidth results in:
- (A) Lower data transfer rate
 - (B) Higher data transfer rate
 - (C) No effect on data rate
 - (D) Increased noise
27. Bandwidth is measured in:
- (A) bps
 - (B) Bytes
 - (C) Hertz
 - (D) Volts
28. Bandwidth is the difference between:
- (A) Signal strength and noise
 - (B) Maximum and minimum frequency
 - (C) Data rate and latency
 - (D) Input and output
29. 1 KHz equals:
- (A) 100 Hz
 - (B) 1000 Hz
 - (C) 1024 Hz
 - (D) 1 MHz
30. 1 MHz equals:
- (A) 1000 Hz
 - (B) 1000 KHz
 - (C) 1024 KHz
 - (D) 10000 Hz
-

Data Transfer Rate

31. Data transfer rate is also called:
- (A) Frequency
 - (B) Bandwidth

- (C) Bit rate
 - (D) Signal rate
32. Data transfer rate measures:
- (A) Characters per second
 - (B) Packets per second
 - (C) Bits per second
 - (D) Frames per second
33. Data transfer rate is measured in:
- (A) Hz
 - (B) bps
 - (C) Bytes
 - (D) Joules
34. 1 Kbps equals:
- (A) 1000 bps
 - (B) 1024 bps
 - (C) 2048 bps
 - (D) 512 bps
35. 1 Mbps equals:
- (A) 1024 Kbps
 - (B) 1000 Kbps
 - (C) 2048 Kbps
 - (D) 512 Kbps
36. 1 Gbps equals:
- (A) 1024 Mbps
 - (B) 1000 Mbps
 - (C) 512 Mbps
 - (D) 2048 Mbps
37. MBps stands for:
- (A) Megabit per second
 - (B) Megabyte per second
 - (C) Microbyte per second
 - (D) Megaband per second
38. Mbps stands for:
- (A) Megabyte per second
 - (B) Megabit per second
 - (C) Microbit per second
 - (D) Mega packet per second

Types of Data Communication

39. Data communication occurs in the form of:
- (A) Packets
 - (B) Signals

- (C) Frames
 - (D) Files
40. Communication channels can be:
- (A) Point-to-point only
 - (B) Multipoint only
 - (C) Point-to-point or multipoint
 - (D) Broadcast only
41. Types of data communication are classified into:
- (A) Two
 - (B) Three
 - (C) Four
 - (D) Five
42. Which of the following is **NOT** a type of data communication?
- (A) Simplex
 - (B) Half-duplex
 - (C) Full-duplex
 - (D) Multicast
-

Simplex Communication

43. Simplex communication is:
- (A) Two-way
 - (B) One-way
 - (C) Multi-way
 - (D) Bidirectional simultaneous
44. In simplex communication:
- (A) Sender and receiver exchange roles
 - (B) Only sender transmits
 - (C) Only receiver transmits
 - (D) Both transmit at same time
45. Which example represents simplex communication?
- (A) Telephone call
 - (B) Walkie-talkie
 - (C) Keyboard input
 - (D) Video call
46. In simplex mode, the entire capacity of the link is used by:
- (A) Receiver
 - (B) Sender
 - (C) Both
 - (D) Network
-

Half-Duplex Communication

47. Half-duplex communication allows:
- (A) One-way transmission
 - (B) Two-way simultaneous transmission
 - (C) Two-way transmission but not simultaneously
 - (D) Broadcast transmission
48. Walkie-talkie is an example of:
- (A) Simplex
 - (B) Half-duplex
 - (C) Full-duplex
 - (D) Multicast
49. In half-duplex communication:
- (A) Sender and receiver transmit together
 - (B) Transmission direction can be switched
 - (C) Only sender transmits
 - (D) Only receiver transmits
-

Full-Duplex Communication

50. Full-duplex communication allows:
- (A) One-direction transmission
 - (B) Transmission at different times
 - (C) Simultaneous transmission in both directions
 - (D) Packet-based transmission
51. Mobile phones use which type of communication?
- (A) Simplex
 - (B) Half-duplex
 - (C) Full-duplex
 - (D) Broadcast
52. Full-duplex communication can be achieved using:
- (A) Single simplex line only
 - (B) Two simplex lines
 - (C) Only wireless medium
 - (D) Only optical fiber

Switching Techniques

53. Switching techniques are mainly used in networks to:
- (A) Increase bandwidth
 - (B) Establish one-to-one communication
 - (C) Reduce signal noise
 - (D) Store transmitted data
54. Making a dedicated connection between every pair of devices is known as:
- (A) Star topology
 - (B) Mesh topology

- (C) Bus topology
 - (D) Ring topology
55. Which topology connects every device to a central device?
- (A) Mesh
 - (B) Ring
 - (C) Star
 - (D) Bus
56. Dedicated connections become costly mainly in:
- (A) Small networks
 - (B) Wireless networks
 - (C) Large networks
 - (D) Optical networks
57. Switching allows data to be:
- (A) Stored permanently
 - (B) Routed through various nodes
 - (C) Encrypted automatically
 - (D) Compressed
58. Switching creates:
- (A) Permanent routes
 - (B) Temporary routes
 - (C) Virtual memory
 - (D) Static paths
59. Which of the following are commonly used switching techniques?
- (A) Time switching and space switching
 - (B) Packet switching and frame switching
 - (C) Circuit switching and packet switching
 - (D) Signal switching and data switching
-

Circuit Switching

60. In circuit switching, a dedicated path is established:
- (A) After data transmission
 - (B) During data transmission
 - (C) Before communication starts
 - (D) After packets arrive
61. Circuit switching uses:
- (A) Multiple paths for packets
 - (B) Same path for all packets
 - (C) Random routing
 - (D) No routing
62. The dedicated path in circuit switching is a sequence of:
- (A) Files
 - (B) Signals

- (C) Links between network nodes
- (D) Frames

63. Traditional telephone systems are an example of:

- (A) Packet switching
- (B) Circuit switching
- (C) Message switching
- (D) Frame relay

64. In circuit switching, the channel remains occupied:

- (A) Only during packet transfer
- (B) Until communication ends
- (C) Only during setup
- (D) Randomly

Packet Switching

65. In packet switching, a message is divided into:

- (A) Frames
- (B) Signals
- (C) Packets
- (D) Bytes

66. Packets in packet switching are transmitted:

- (A) Together
- (B) Independently
- (C) Sequentially only
- (D) Using a fixed path

67. Different packets of the same message may:

- (A) Take the same route always
- (B) Take different routes
- (C) Be dropped permanently
- (D) Be encrypted differently

68. Each packet contains:

- (A) Only message
- (B) Only header
- (C) Header and message
- (D) Footer only

69. The header of a packet contains:

- (A) Data content
- (B) Destination address
- (C) Encryption key
- (D) File size

70. In packet switching, the channel is occupied:

- (A) Permanently
- (B) Only during packet transmission

- (C) Only during setup
 - (D) Only during reception
71. After packet transmission, the channel becomes:
- (A) Blocked
 - (B) Reserved
 - (C) Available for others
 - (D) Damaged
72. VoIP is used for delivering:
- (A) Only voice
 - (B) Only video
 - (C) Voice and multimedia over IP
 - (D) Only text
73. VoLTE is related to:
- (A) Satellite communication
 - (B) Wired telephony
 - (C) High-speed wireless communication
 - (D) Optical fiber
-

Transmission Media

74. Transmission media refers to:
- (A) Software
 - (B) Hardware devices
 - (C) Path carrying signals
 - (D) Protocols
75. Electric wire carrying current to a fan is an example of:
- (A) Signal
 - (B) Medium
 - (C) Node
 - (D) Protocol
76. In human communication, air acts as:
- (A) Signal
 - (B) Receiver
 - (C) Medium
 - (D) Protocol
77. Transmission media in data communication carry:
- (A) Programs
 - (B) Messages
 - (C) Instructions
 - (D) Algorithms
78. Transmission media are classified into:
- (A) Analog and digital
 - (B) Electrical and optical

- (C) Guided and unguided
 - (D) Fast and slow
-

Guided (Wired) Media

79. Guided transmission uses:
- (A) Air
 - (B) Antenna
 - (C) Physical cables
 - (D) Satellites
80. Wired media are also known as:
- (A) Unguided media
 - (B) Optical media
 - (C) Guided media
 - (D) Virtual media
81. Which of the following is NOT a wired medium?
- (A) Twisted pair
 - (B) Coaxial cable
 - (C) Optical fiber
 - (D) Radio waves
-

Twisted Pair Cable

82. A twisted pair cable consists of:
- (A) One copper wire
 - (B) Two twisted copper wires
 - (C) Glass fiber
 - (D) Aluminum wire
83. Twisting of wires helps in:
- (A) Increasing weight
 - (B) Reducing interference
 - (C) Increasing length
 - (D) Reducing bandwidth
84. Twisted pair cables are commonly used in:
- (A) Satellite links
 - (B) Telephone lines and LANs
 - (C) Backbone networks
 - (D) Microwave communication
85. Types of twisted pair cables are:
- (A) Thick and thin
 - (B) Copper and aluminum
 - (C) UTP and STP
 - (D) Analog and digital

86. STP stands for:
- (A) Standard Twisted Pair
 - (B) Shielded Twisted Pair
 - (C) Simple Twisted Pair
 - (D) Secure Transmission Pair
-

Coaxial Cable

87. Coaxial cable has higher bandwidth than:
- (A) Optical fiber
 - (B) Twisted pair
 - (C) Radio waves
 - (D) Infrared
88. The core of a coaxial cable is made of:
- (A) Plastic
 - (B) Glass
 - (C) Copper wire
 - (D) Aluminum foil
89. The shielding in coaxial cable helps in:
- (A) Cooling
 - (B) Reducing interference
 - (C) Increasing resistance
 - (D) Data encryption
90. Coaxial cables are suitable for:
- (A) Short distance, low frequency
 - (B) Long distance, high frequency
 - (C) Wireless transmission
 - (D) Infrared transmission
-

Optical Fiber Cable

91. Optical fiber transmits data in the form of:
- (A) Electrical signals
 - (B) Radio waves
 - (C) Light
 - (D) Infrared heat
92. Optical fiber uses which principle to guide light?
- (A) Reflection
 - (B) Refraction
 - (C) Diffraction
 - (D) Dispersion
93. Optical fiber cables are commonly used in:
- (A) LAN only
 - (B) Home wiring

- (C) Backbone networks
 - (D) Telephone handsets
94. Optical fibers are:
- (A) Heavy and slow
 - (B) Lightweight with high bandwidth
 - (C) Low bandwidth
 - (D) Easily affected by noise
95. Optical fiber cables are:
- (A) Bidirectional by default
 - (B) Unidirectional
 - (C) Wireless
 - (D) Analog only
96. Full-duplex communication using optical fiber requires:
- (A) One cable
 - (B) Two cables
 - (C) No cables
 - (D) Satellite
-

Wireless Transmission Media

97. Wireless transmission uses:
- (A) Electric current
 - (B) Light signals
 - (C) Electromagnetic waves
 - (D) Sound waves
98. Wireless media are also called:
- (A) Guided media
 - (B) Wired media
 - (C) Unguided media
 - (D) Optical media
99. The electromagnetic spectrum used for wireless communication ranges from:
- (A) 3 Hz to 300 Hz
 - (B) 3 KHz to 900 THz
 - (C) 300 MHz to 1 GHz
 - (D) 1 GHz to 300 GHz
100. Dish-shaped antennas are used for:
- (A) Short distance communication
 - (B) Long distance communication
 - (C) Optical communication
 - (D) Wired networks
101. Repeaters are used to:
- (A) Block signals
 - (B) Regenerate signals

- (C) Encrypt data
 - (D) Compress packets
-

Electromagnetic Waves

102. The electromagnetic spectrum is divided into:
- (A) Two categories
 - (B) Three categories
 - (C) Four categories
 - (D) Five categories
103. Which of the following has the lowest frequency range?
- (A) Infrared waves
 - (B) Microwaves
 - (C) Radio waves
 - (D) Light waves
104. Radio waves are:
- (A) Unidirectional
 - (B) Omni-directional
 - (C) Optical
 - (D) Infrared
105. Microwaves require:
- (A) Reflection
 - (B) Line-of-sight
 - (C) Diffraction
 - (D) Refraction
106. Infrared waves are mainly used for:
- (A) Long distance communication
 - (B) Satellite communication
 - (C) Short-range communication
 - (D) Broadcasting
-

Bluetooth Technology

107. Bluetooth is a:
- (A) Long-range technology
 - (B) Short-range wireless technology
 - (C) Wired protocol
 - (D) Optical technology
108. Bluetooth operates in which frequency band?
- (A) 900 MHz
 - (B) 1.8 GHz
 - (C) 2.4 GHz
 - (D) 5 GHz

109. Bluetooth range is approximately:
- (A) 1 meter
 - (B) 5 meters
 - (C) 10 meters
 - (D) 100 meters
110. Bluetooth data transfer speed is:
- (A) 100 Mbps
 - (B) 10 Mbps
 - (C) 1–2 Mbps
 - (D) 1 Gbps
111. A Bluetooth personal area network is called:
- (A) WLAN
 - (B) MAN
 - (C) Piconet
 - (D) Internet
112. In a piconet, devices work in:
- (A) Peer-to-peer
 - (B) Client-server
 - (C) Master-slave
 - (D) Broadcast
113. A Bluetooth master can communicate with how many active slave devices?
- (A) 5
 - (B) 7
 - (C) 8
 - (D) 255

Wireless LAN (Wi-Fi)

114. Wireless LAN is an example of:
- (A) WAN
 - (B) MAN
 - (C) LAN
 - (D) PAN
115. Wireless LAN is popularly known as:
- (A) Bluetooth
 - (B) WiMAX
 - (C) Wi-Fi
 - (D) LTE
116. IEEE standard number assigned to wireless LAN is:
- (A) 802.3
 - (B) 802.5
 - (C) 802.11
 - (D) 802.15
117. Devices used in WLAN include:
- (A) Only servers

- (B) Only routers
 - (C) Laptops and mobile phones
 - (D) Printers only
118. Access Point (AP) is used to:
- (A) Encrypt data
 - (B) Create a wireless LAN
 - (C) Store web pages
 - (D) Process signals
119. An access point connects to:
- (A) Wireless router only
 - (B) Antenna only
 - (C) Wired network
 - (D) Satellite
120. Devices in a WLAN access the Internet through:
- (A) Switch
 - (B) Hub
 - (C) Access Point
 - (D) Modem only
121. One advantage of WLAN is:
- (A) High installation cost
 - (B) No mobility
 - (C) Increased access for mobile devices
 - (D) Limited coverage
122. WLAN helps in:
- (A) Replacing wired infrastructure
 - (B) Reducing bandwidth
 - (C) Increasing interference
 - (D) Slowing data transfer

Mobile Telecommunication Technologies

123. The most widely used network today is:
- (A) LAN
 - (B) MAN
 - (C) WAN
 - (D) Mobile phone network
124. Mobile networks allow communication:
- (A) Only while stationary
 - (B) Only in offices
 - (C) On-the-go
 - (D) Only at home
125. Internet of Things (IoT) enables:
- (A) Manual device control

- (B) Communication between smart devices
- (C) Only wired communication
- (D) Data deletion

126. Mobile communication technologies are classified based on:

- (A) Speed
- (B) Architecture
- (C) Cost
- (D) Frequency

127. Mobile generations are identified as:

- (A) 1A, 2A, 3A
- (B) G1, G2, G3
- (C) 1G, 2G, 3G, 4G, 5G
- (D) A, B, C

1G Mobile Technology

128. The first generation mobile network was introduced around:

- (A) 1972
- (B) 1982
- (C) 1992
- (D) 2002

129. 1G technology was used for:

- (A) Voice only
- (B) Data only
- (C) Voice and data
- (D) Multimedia

130. 1G used which type of signals?

- (A) Digital
- (B) Optical
- (C) Analog
- (D) Infrared

2G Mobile Technology

131. 2G mobile networks were introduced around:

- (A) 1981
- (B) 1986
- (C) 1991
- (D) 1996

132. 2G transmitted voice calls in:

- (A) Analog form
- (B) Optical form
- (C) Digital form
- (D) Infrared form

133. Compared to 1G, 2G provided:
- (A) Lower quality
 - (B) Improved call quality
 - (C) No security
 - (D) Slower transmission
134. 2G enabled which additional services?
- (A) Email only
 - (B) SMS and MMS
 - (C) Video calling
 - (D) Internet browsing
-

3G Mobile Technology

135. 3G was commercially introduced around:
- (A) 1995
 - (B) 1998
 - (C) 2001
 - (D) 2005
136. 3G supports:
- (A) Analog voice only
 - (B) Digital voice only
 - (C) Digital voice and data
 - (D) Optical communication
137. Internet access in 3G is provided via:
- (A) Wired routers
 - (B) Optical fiber
 - (C) Same radio towers as voice
 - (D) Satellites only
138. 3G allowed:
- (A) Fewer simultaneous calls
 - (B) Same frequency for fewer users
 - (C) More simultaneous calls
 - (D) No data transmission
-

4G Mobile Technology

139. 4G was developed mainly to:
- (A) Reduce call quality
 - (B) Increase voice delay
 - (C) Support faster data
 - (D) Remove multimedia
140. Compared to 3G, 4G is:
- (A) Slower
 - (B) Same speed

- (C) Much faster
- (D) Less efficient

141. 4G supports:

- (A) Text only
- (B) Voice only
- (C) Interactive multimedia and broadband
- (D) Analog signals

142. Technologically, 4G is:

- (A) Same as 3G
- (B) Slightly different
- (C) Completely different
- (D) Older than 3G

5G Mobile Technology

143. 5G is expected to support:

- (A) Only voice
- (B) Only SMS
- (C) IoT and M2M communication
- (D) Analog data

144. Machine to Machine (M2M) communication means:

- (A) Human to machine
- (B) Device to device
- (C) Human to human
- (D) Network to network

145. 5G data transfer speed is expected in:

- (A) Kbps
- (B) Mbps
- (C) Gbps
- (D) Tbps

146. 5G is expected to support:

- (A) Only smartphones
- (B) Connected vehicles and IoT devices
- (C) Only computers
- (D) Only landlines

Protocol

147. Protocol refers to:

- (A) Hardware devices
- (B) Transmission media
- (C) Set of standard rules
- (D) Signal strength

148. Protocol rules must be followed by:
- (A) Only sender
 - (B) Only receiver
 - (C) Only routers
 - (D) Sender, receiver, and intermediate devices
149. Different networks may require:
- (A) Same data rate always
 - (B) Same format always
 - (C) Different data formats
 - (D) No protocol
-

Need for Protocols

150. Flow control is required when:
- (A) Both devices have same speed
 - (B) Sender is slower
 - (C) Sender and receiver have different speeds
 - (D) Network is idle
151. Access control is required to:
- (A) Increase bandwidth
 - (B) Avoid packet collision
 - (C) Encrypt data
 - (D) Compress data
152. Packet collision leads to:
- (A) Faster transmission
 - (B) Data loss or corruption
 - (C) Higher bandwidth
 - (D) Better security
153. Protocols help in:
- (A) Identifying computers
 - (B) Data formatting
 - (C) Packet reassembly
 - (D) All of the above
-

HyperText Transfer Protocol (HTTP)

154. HTTP stands for:
- (A) Hyper Transfer Text Protocol
 - (B) HyperText Transfer Protocol
 - (C) High Transfer Text Protocol
 - (D) Host Transfer Text Protocol
155. HTTP is mainly used to access:
- (A) FTP servers
 - (B) Email servers

- (C) World Wide Web
 - (D) LAN
156. HTTP follows which model?
- (A) Peer-to-peer
 - (B) Client-server
 - (C) Broadcast
 - (D) Multicast
157. HTTP runs over:
- (A) UDP
 - (B) IP only
 - (C) TCP
 - (D) ICMP
158. HTTP was developed at:
- (A) NASA
 - (B) CERN
 - (C) Microsoft
 - (D) Google
159. HTTP was developed in:
- (A) 1980
 - (B) 1985
 - (C) 1989
 - (D) 1995
160. A web page is written using:
- (A) C++
 - (B) Java
 - (C) HTML
 - (D) Python
161. URL stands for:
- (A) Uniform Resource Locator
 - (B) Universal Resource Link
 - (C) Unified Routing Link
 - (D) Uniform Routing Locator
-

File Transfer Protocol (FTP)

162. FTP is used for:
- (A) Email transfer
 - (B) Web browsing
 - (C) File transfer
 - (D) Chatting
163. FTP works on:
- (A) Peer-to-peer model
 - (B) Client-server model

- (C) Broadcast model
- (D) Ring model

164. FTP may require:

- (A) Authentication
- (B) Encryption always
- (C) Satellite link
- (D) Optical cable

165. FTP handles issues related to:

- (A) File naming conventions
- (B) Directory structures
- (C) Data representation
- (D) All of the above

Point-to-Point Protocol (PPP)

166. PPP establishes:

- (A) Multipoint connection
- (B) Dedicated direct connection
- (C) Broadcast communication
- (D) Wireless LAN

167. PPP is used between:

- (A) Two routers
- (B) Two switches
- (C) Two printers
- (D) Two hubs

168. PPP ensures:

- (A) Packet loss
- (B) Data integrity
- (C) Signal noise
- (D) Encryption only

169. PPP requires devices to have:

- (A) Simplex mode
- (B) Half-duplex mode
- (C) Duplex mode
- (D) Broadcast mode

Simple Mail Transfer Protocol (SMTP)

170. SMTP is used for:

- (A) File transfer
- (B) Email services
- (C) Web access
- (D) Chat applications

171. SMTP is concerned with:
- (A) Email content
 - (B) Email encryption
 - (C) Message header
 - (D) Image attachments
172. SMTP sender program takes mails from:
- (A) Inbox
 - (B) Spam
 - (C) Outgoing queue
 - (D) Drafts
173. SMTP receiver places received mail in:
- (A) Server memory
 - (B) Cache
 - (C) User mailbox
 - (D) Router
-

TCP / IP

174. TCP/IP is a:
- (A) Single protocol
 - (B) Hardware device
 - (C) Set of protocols
 - (D) Transmission media
175. IP ensures:
- (A) Packet ordering
 - (B) Unique identification of nodes
 - (C) Data encryption
 - (D) Compression
176. TCP is responsible for:
- (A) Assigning IP addresses
 - (B) Breaking data into packets
 - (C) Physical transmission
 - (D) Signal modulation
177. TCP guarantees:
- (A) Packet loss
 - (B) Packet delivery
 - (C) Signal noise
 - (D) Data corruption
178. TCP ensures packets are delivered:
- (A) Randomly
 - (B) In reverse order
 - (C) In correct sequence
 - (D) Without headers

179. Different packets of same message may:

- (A) Follow same route always
- (B) Follow different routes
- (C) Never reach destination
- (D) Be discarded

180. At the destination, packets are:

- (A) Destroyed
- (B) Stored permanently
- (C) Reassembled
- (D) Encrypted

Numerical & Summary-Based MCQs

181. Bandwidth is measured in:

- (A) bps
- (B) Hz
- (C) Bytes
- (D) Volts

182. Data transfer rate is measured in:

- (A) Hz
- (B) Bits per second
- (C) Joules
- (D) Amperes

183. Optical fiber is immune to:

- (A) Heat
- (B) Noise
- (C) Light
- (D) Refraction

184. Which communication mode uses entire capacity in one direction?

- (A) Full-duplex
- (B) Half-duplex
- (C) Simplex
- (D) Broadcast

185. Which switching technique breaks messages into packets?

- (A) Circuit switching
- (B) Packet switching
- (C) Star switching
- (D) Bus switching